



Nokia E51 phone with Internet browser

possibilities.

"Saying that a BMW M5 would support Facebook and Twitter integration is an indication that we are getting there," quips Sridharan Mani, director and CEO, American Megatrends India. M2M World News Insight says that in 2010 around 22 million connected consumer devices were shipped with the number likely to double in 2011. Forecasts indicate that by 2015 around 270 million consumer devices with embedded connectivity would be shipped.

If you set aside typical computers and look at the embedded Internet market today, it is very evident that mobile telephony is the apple of the industry's eye. "Embedded devices catering to the mobile segment, like iPhones or Android phones, are most equipped today owing to the widespread adoption of 3G, 4G LTE and other high-speed data networks. With the advent of latest tablet PCs and e-book readers with Internet connectivity using 3G modems, or with Wi-Fi, the mobile telephony segment has the most widespread embedded capability today. This trend is most likely to increase in the future with mobile Internet overtaking desktop Internet, just like how mobiles now outnumber landlines," explains Mani.

Other consumer electronic (CE) devices such as televisions are also beginning to get connected. Even the averagely-priced models of modern

televisions have an ethernet port to access content from the Internet. Similarly, brands like Canon and Samsung also have digital cameras that can connect to the Internet using a Wi-Fi connection, and directly upload pictures to popular photo-sharing websites. There are also video cameras that can stream movies live over the Internet. These are still a preserve of the elite though.

Anup Sable, senior vice president-automotive and engineering, KPIT Cummins feels that household CE devices like kitchenettes, washing machines and ovens will also have connectivity as a key feature in the near future.

Why would you want to have an oven or washing machine that connects to the Internet? Imagine the convenience of being able to 'connect' to your oven and set it to start cooking just before you leave from office. Hot dinner ready in a jiffy! If something goes wrong with your washing machine the service person would be able to 'connect' to it from his office, diagnose the problem, and perhaps even set it right from there; or at least come equipped with the parts needed to fix the problem during his first visit itself.

Cars are also getting more Internet-savvy. Although automobiles have long been able to connect to networks—mainly GPS applications—more generalised Internet capabilities are also being talked about now. A number of Internet radios

and in-car modems are being introduced in the market; and the latest BMW supports an iPhone application for Facebook and Twitter integration.

Industrial automation and home automation systems are also a huge market for the embedded Internet industry. Internet-capable measurement instruments as well as machinery are used extensively today—to feed data live to centralised IT solutions, for monitoring and adjusting processes as well as for decision-making. Likewise, energy management systems, intelligent lighting solutions and premise security applications utilise a lot of Internet-capable devices. Some CCTV camera models, for example, come with Internet capability, and the output of these can be viewed from any remote PC or mobile phone.

Prof. Anupama remarks, "The trend will move towards military and industrial applications where embedded devices will begin to meet more and more real-time demands. After a certain period of advancement, the trend will move more towards implementing the embedded devices for day-to-day needs like traffic monitoring, precision agriculture, water resources management, resource mining, etc."

Basic components—not too different

Like all computers, embedded systems also comprise processors, operating systems and software stacks—just that they use small and barebones versions of everything, so that the footprint in terms of size, power, etc, is small enough to be embedded into another device. To add Internet-capability, the system would also have to include a modem and the relevant software stacks to establish connection. Depending on the application, other software or hardware components might be added. For example, a mobile phone might also include e-mail clients, browsers, special purpose players, etc.

As far as the Internet capability goes, again the components draw a parallel